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SYSTEMATIC REVIEW

Relationships Between Training Load Indicators and Training Outcomes in Professional Soccer

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Abstract

Background In professional senior soccer, training load monitoring is used to ensure an optimal workload to maximize physical fitness and prevent injury or illness. However, to date, different training load indicators are used without a clear link to training outcomes.

Objective The aim of this systematic review was to identify the state of knowledge with respect to the relationship between training load indicators and training outcomes in terms of physical fitness, injury, and illness.

Methods A systematic search was conducted in four electronic databases (CINAHL, PubMed, SPORTDiscus, and Web of Science). Training load was defined as the amount of stress over a minimum of two training sessions or matches, quantified in either external (e.g., duration, distance covered) or internal load (e.g., heart rate [HR]), to obtain a training outcome over time.

Results A total of 6492 records were retrieved, of which 3304 were duplicates. After screening the titles, abstracts and full texts, we identified 12 full-text articles that matched our inclusion criteria. One of these articles was identified through additional sources. All of these articles used correlations to examine the relationship between load indicators and training outcomes. For pre-season, training time spent at high intensity (i.e., >90 % of maximal HR) was linked to positive changes in aerobic fitness. Exposure time in terms of accumulated training, match or combined training, and match time showed both positive and negative relationships with changes in fitness over a season. Muscular perceived exertion may indicate negative changes in physical fitness. Additionally, it appeared that training at high intensity may involve a higher injury risk. Detailed external load indicators, using electronic performance and tracking systems, are relatively unexamined. In addition, most research focused on the relationship between training load indicators and changes in physical fitness, but less on injury and illness.

Conclusion HR indicators showed relationships with positive changes in physical fitness during pre-season. In addition, exposure time appeared to be related to positive and negative changes in physical fitness. Despite the availability of more detailed training load indicators nowadays, the evidence about the usefulness in relation to training outcomes is rare. Future research should implement continuous monitoring of training load, combined with the individual characteristics, to further examine their relationship with physical fitness, injury, and illness.

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Key Points

Inconsistent findings exist regarding relationships between external load indicators and changes in sprint and jump performance in professional senior soccer.

There is moderate evidence for relationships between heart rate-based training load and changes in aerobic fitness during pre-season.

Despite the availability of external and internal load indicators, evidence regarding their relevance in relation to injury or illness is scarce.

1 Introduction

General objectives of physical training in professional senior soccer are the ability to compete at the highest level possible throughout the season and to prevent injuries. Therefore, a well-planned progressive build-up in training with variations in frequency, duration, intensity, and type of activities should be respected. It is assumed that combinations of training stimuli and sufficient recovery will increase physical fitness [1]. In addition, training load quantification is considered crucial for accurate training prescription and evaluation, which subsequently leads to potential improvements in physical fitness [2]. On the downside, inappropriate training stimuli or too little recovery may result in a lack of fitness or an increased risk for injury and illness. Taken together, the training process can result in different outcomes: performance gains (e.g., improved physical fitness) or performance losses (e.g., injury or illness) [3].

In professional soccer, the global burden of injuries consists of an average of 2.0 injuries per player per season and an injury incidence of 8.0 injuries per 1000 h of training and match exposure [4]. In particular, muscle injuries (e.g., hamstring and calf) were found to be a considerable problem for players and clubs, as they cause almost one-third of all time-loss injuries [5]. With respect to illnesses, infections mostly concern the upper respiratory or digestive system; however, their impact is small compared with injuries in general and muscle injuries in particular, as only 19.7 % result in training or match absence [6]. Possibly, a combination of the increased demands of the game [7] and congested fixtures [8, 9] together with mismatches between load and recovery processes underlie these injury rates [10]. This stresses the need for the continuous monitoring of load and recovery to prevent injury and illness.

The process of continuous monitoring could lead to better insight into the actual training load and potentially result in changes to training prescription in subsequent sessions (i.e., increase or decrease load). Training load can be differentiated into external and internal loads [11]. The external load refers to the overall activities of players, often imposed by the coach (e.g., all of the movement actions during 5 × 3 min of small-sided games with a 30 s rest interval). The internal load is the actual physiological and psychological stress imposed on the player's body. This type of load is considered the ultimate stimulus for training adaptations [12] and results from the interaction between the external load and the individual or intrinsic characteristics of a player [13]. These characteristics are intrinsic factors such as training history, actual fitness, body composition, strength asymmetries, and injury history [14]. Due to these individual differences, a given external load can elicit a different internal load for each player within a team.

To date, different parameters have been used to quantify the external and internal load [15]. Different methods using heart rate (HR) have been examined in individual sports and team sports. For example, the training impulse (TRIMP) is an HR-based method, converting the HR data from a training session into an arbitrary unit (AU) that reflects the physical effort. Another method used to study internal workload consists of multiplying the session rating of perceived exertion (RPE) by session duration [2], which is well-known as training load (TL_{RPE}). In addition, some variations are examined such as weekly TL_{RPE}, weekly monotony (i.e., the daily mean of weekly TL_{RPE} divided by the standard deviation), and weekly strain (i.e., the total weekly TL_{RPE} multiplied by monotony) [16]. For external load parameters, power output measured by ergometers for cycling [17] and time–motion measures for running [18] are used in individual sports. In team sports, time–motion analysis together with accelerometry are implemented [19]. These relatively new technological developments provide the opportunity to quantify external load for the individual players in a valid and reliable way [20, 21].

Both external and internal load parameters can be used to model and possibly predict the effect of training load on performance [22–25]. To examine the link between training load and performance, various models have been presented that consider the athlete as an input–output system [22]. In the fitness–fatigue model [26, 27], for example, training load acts as the dose and performance as the response. Specifically, it suggests that the predicted performance is the difference between the actual fitness and the residual fatigue. Several predictive models for changes in fitness exist, mostly focusing on endurance sports. Some of them are based on the original fitness–fatigue model [28], others use a metamodel [29] or artificial neural networks [30]. One study connected a high percentage of illnesses in a group of endurance athletes to the exceeding of individually identified training load thresholds [16].

Regardless of the abundance of measurement tools, training load parameters, and research findings from other football codes [31–36] and amateur [37] or youth [38, 39] soccer players, it remains unclear which indicators of load best relate to changes in physical fitness, injury, and illness in professional senior soccer. Therefore, the purpose of this paper was to conduct a systematic review to determine the state of knowledge in professional male soccer regarding the relationship between training load indicators and training outcomes.

2 Methods

2.1 Literature Search Strategy

Studies for this review were identified through a systematic search of four electronic databases: CINAHL, PubMed, SPORTDiscus, and Web of Science. Combinations of the following groups of keywords were used: (i) 'soccer', 'football'; (ii) 'training', 'match*', 'game*', 'competition'; (iii) 'load', 'stress', 'work', 'workload', 'type', 'frequency', 'volume', 'time', 'duration', 'intens*'; (iv) 'perform*', 'fitness', 'capacit*', 'abilit*', 'freshness', 'readiness', 'preparedness'; (v) 'injur*', 'overuse', 'trauma', 'illness', 'overreach*', 'overtrain*', 'burnout', 'risk*', 'accident*', 'fatigue', 'incident*', '*exert*', 'staleness', 'recover*'.

Terms were connected with OR within each group. Two searches on title and abstract were conducted in each database by combining groups (i), (ii), and (iii), with either group (iv) or (v) using AND. The results from the two searches were then combined using OR, similar to the method of Gabbett et al. [40] who focused on all football codes in adolescent players.

The search was restricted to English peer-reviewed journals until 26 March 2015. Due to the fixed combination of title, abstract, author keywords, and KeyWords Plus® in Web of Science, an additional search was carried out using reference management software (EndNote™ X7.1, Thomson Reuters, New York, NY, USA). Again, only articles with 'soccer' or 'football' in the title or abstract were retained to have an analogous search strategy for all electronic databases.

2.2 Selection Criteria

One author initially checked the title and abstract of the articles and excluded all duplicates. Thereafter, two authors checked the title and abstract of studies evaluating professional soccer players regarding the usage of external and internal load indicators and their potential relationship to changes in fitness, injury, or illness. For the purpose of this review, articles with a minimum of two training sessions or matches were included. Articles were excluded if (1) the participants were not professional male soccer players; (2) the participants were players aged under 18 years; (3) the participants were not monitored longitudinally (i.e., over a minimum of two training sessions or matches); (4) the articles did not report any indicators of external or internal load in terms of volume, intensity, or both, in accordance with the classification of Halson [15]; (5) there was no focus on the relationship between load and changes in physical fitness, injury, or illness for individual players (e.g., group effects of a general training program); (6) the articles investigated the effects of rehabilitation, nutritional strategies, or environmental factors (e.g., heat, altitude); or (7) the articles were reviews, case studies, or abstracts.

In case of ambiguity in the abstracts about any of the criteria, we checked the full-text article for verification. Any disagreements were discussed between two authors and resolved. The full-text articles of the remaining studies were then downloaded and archived. These articles were examined using the same criteria.

The references of the selected articles were then checked to identify any potentially relevant articles not identified by the original search. A Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart was used to clarify the study's identification, screening, eligibility, inclusion, and analysis (Fig. 1).

3 Results

After screening the titles, abstracts, and full texts, 11 articles met the criteria for inclusion in the review. One additional article was identified through additional sources (Table 1).

The group sizes of the studies varied between ten and 35 subjects. Nine studies were conducted in the highest professional league in Italy, Spain, France, Portugal, and Scotland, two studies in the second-highest Spanish division, and one study in the third-highest Spanish professional division.

The minimum duration of data collection was one training week [41]. Three studies monitored 6–8 weeks during pre-season [42–44] and one study comprised 5 weeks of pre-season and 4 weeks of in-season [45]. Two studies were completed in-season and lasted 17 days [46] and 9 weeks [47], respectively. Finally, three studies

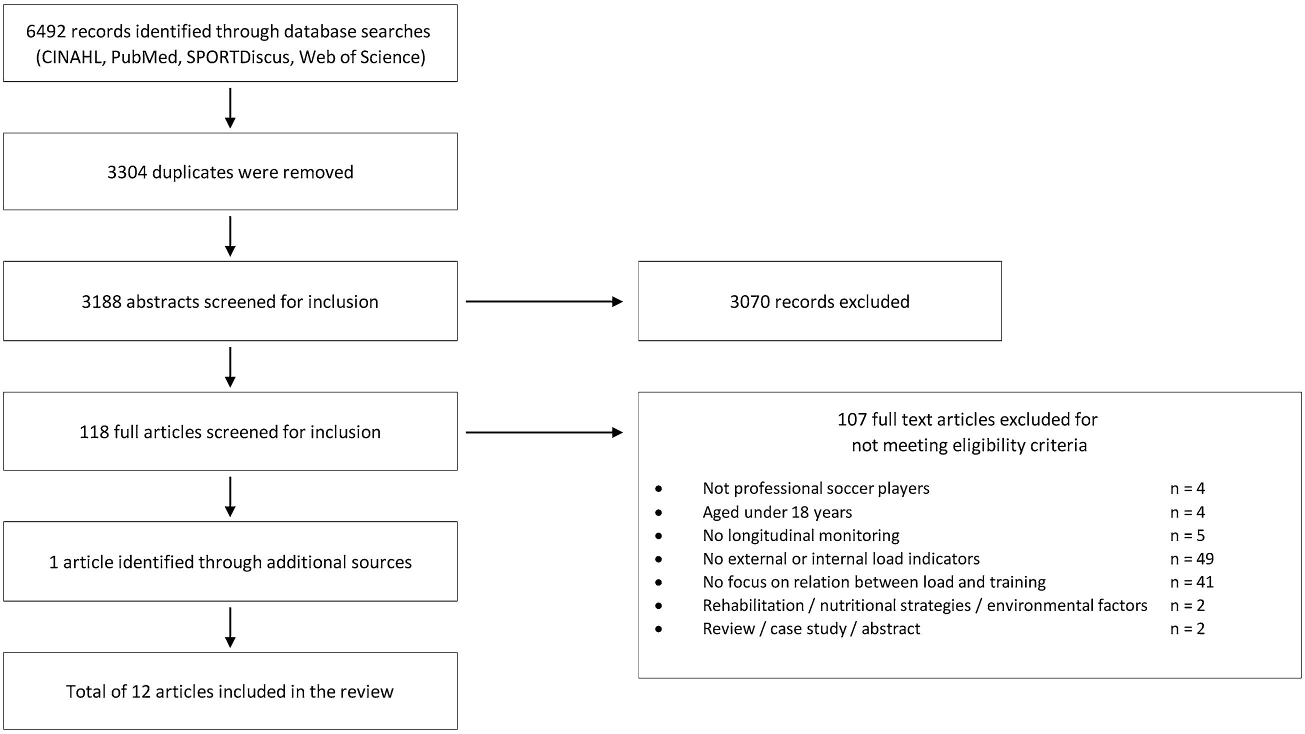

Fig. 1 Preferred reporting items for systematic reviews and meta-analyses (PRISMA) flowchart outlining the selection process for inclusion of articles in the systematic review

focused on one entire season [48–50] and two studies were completed over two consecutive seasons [51, 52].

3.1 Changes in Fitness

3.1.1 Relationship with External Load Indicators

The relationship between external load and changes in fitness has been examined in six studies. Two studies focused on accumulated match time and three studies monitored total exposure time (i.e., combined accumulated training and match time). One study focused on a time–motion parameter.

No significant associations were found between exposure time variables and changes in aerobic fitness. In terms of time–motion variables, the total distance of high-speed running (THSR; 14.4 km·h^{-1}) was measured by means of the global positioning system (GPS). This THSR was significantly correlated ($r = -0.2$) with daily changes in vagal-related HR variability (HRV) [46]. HRV was measured, together with HR recovery (HRR), during an indoor submaximal 5 min cycling/5 min recovery test prior to every training session. Based on the results of a regression analysis, an increase of 300 m in THSR was associated with a decline of 1 unit in HRV, possibly indicating an increased aerobic fitness [53]. There was no significant correlation found for THSR and HRR.

Positive as well as negative relationships were found for exposure time variables and sprint performance. More match playing time correlated significantly with a better performance in sprint testing. It was found that a higher accumulated match time resulted in greater relative improvements in a 5 m sprint [49]. Another study revealed that a higher accumulated match time was related to better 30 m sprint time [50]. For total exposure time, Los Arcos et al. [45] found impaired sprint performances over 5 and 15 m.

For countermovement jump (CMJ) testing, results were inconsistent. A higher accumulated match time was related to improved CMJ performance [49]. In contrast, two studies concluded that a higher total exposure time was negatively related to performance on CMJ [47] as well as in CMJ with arm swing [45]. For THSR, a positive correlation ($r = 0.23$) was found with CMJ [46], indicating that an increased distance covered above 14.4 km·h^{-1} was related to an improved CMJ. In addition, Silva et al. [49] found significant correlations between CMJ and sprint times over 5 and 30 m. Correlations fluctuated between –0.47 and –0.73, indicating significant relationships between sprint performance and CMJ. This was in accordance with Haugen et al. [54], who found significant correlations in anaerobic testing for sprinting velocity over 20 m and CMJ.

Two studies showed beneficial changes following higher accumulated match time for isokinetic leg strength

Table 1 Summary of results from studies investigating the relationship between training load indicators and training outcomes in professional soccer players

Study (year)	Number (n), age [mean (SD)], body height and mass [mean (SD)] of participants	External load indicators	Internal load indicators	Physiological assessment and training outcome, including injuries	Duration	Relationships between load indicators and training outcomes
Carling and Orhant [48] (2010)	30 professional players, 24.4 (±4.1) years, 182.1 (±5.8) cm, 76.8 (±5.8) kg	Total exposure time	No internal load measures	Body composition measures	1 full season	No significant relationships between total exposure time and estimated body fat, body mass, and fat-free mass
Castagna et al. [42] (2011)	14 professional players, 25 (±4) years, 178 (±7) cm, 74 (±8) kg	No external load measures	HR	S2 and S4 during incremental treadmill testing, submaximal aerobic fitness assessment	6 weeks of pre-season	Training time spent at HI was significantly related to relative speed improvements in S2 and S4 during incremental testing
Castagna et al. [43] (2013)	18 professional players, 28.6 (±3.2) years, 183 (±6.1) cm, 80.0 (±5.4) kg	No external load measures	HR	S2 and S4 during incremental treadmill testing, $\dot{V}O_{2max}$ test, YYIR1	8 weeks of pre-season	Training time spent at HI was significantly related to relative speed improvements in S2 and S4 during incremental testing, as well as to improvements in $\dot{V}O_{2max}$ and YYIR1
Los Arcos et al. [47] (2014)	21 professional players, 21.0 (±1.7) years, 181.0 (±6.3) cm, 76.1 (±7.7) kg	Total exposure time	RPE_{mus}, RPE_{res}, TL_{RPEmus}, TL_{RPEres}, and HR	Body mass, CMJ, 15 m sprint test (5 and 15 m sprint times), submaximal constant velocity endurance running test	9 weeks of in-season	Significant negative correlations between values of total exposure time and relative changes in CMJ, as well as between total added RPE_{mus} and relative changes in 15 m sprinting speed
Los Arcos et al. [45] (2015)	14 professional players, 20.6 (±1.7) years, 179.0 (±6.0) cm, 73.5 (±7.0) kg	Total exposure time	RPE_{mus}, RPE_{res}, TL_{RPEmus}, TL_{RPEres}, and HR	CMJ, CMJ_{as}, CMJ_{dl}, CMJ_{ndl}, 15 m sprint test (5 and 15 m sprint times), submaximal aerobic fitness running test	9 weeks, including 5 weeks of pre-season and 4 weeks of in-season	Significant negative correlations between total exposure time and relative changes in CMJ_{as} and 5 and 15 m sprint times, between both sum of RPE_{mus} and TL_{RPEmus} and relative changes in CMJ_{dl} and CMJ_{ndl}, and between TL_{RPEmus} and changes in blood lactate at 13 km·h^{-1} in submaximal testing
Mallo and Dellal [52] (2012)	35 professional players, 21.4 (±2.4) years, 180.2 (±5.5) cm, 74.0 (±5.1) kg	Total exposure time	HR	Injury data collection according to methods of Hägglund et al. [70] and Fuller et al. [71]	2 full seasons	Incidence of muscle strains was significantly related to mean HR during training stage, but not to training frequency or training volume
Manzi et al. [44] (2013)	18 professional players, 28.4 (±3.2) years, 182.0 (±5.3) cm, 79.9 (±5.5) kg	No external load measures	$TRIMP_i$	S4 during incremental treadmill testing, $\dot{V}O_{2max}$, VT, and $\dot{V}O_2$ at VT using progressive maximal treadmill testing, YYIR1	8 weeks of pre-season	Significant positive relationships between $TRIMP_i$ and changes in $\dot{V}O_{2max}$, VT, S4, and YYIR1 A weekly $TRIMP_i$ >500 AU was necessary to ensure improvements in pre-season

Table 1 continued

Study (year)	Number (n), age [mean (SD)], body height and mass [mean (SD)] of participants	External load indicators	Internal load indicators	Physiological assessment and training outcome, including injuries	Duration	Relationships between load indicators and training outcomes
Owen et al. [41] (2014)	10 professional players, 26.8 (±4.1) years, 185.4 (±6.0) cm, 79.3 (±7.8) kg	Total exposure time, TD, THSR (>21.6 km·h^{-1}), FEHS (>21.6 km·h^{-1}), %HSR, m·min^{-1}	HR, RPE, and TL_{RPE}	Daily wellness questionnaire, s-IgA	1 training week during in-season	Significant negative relationships were observed for HI training sessions: (i) between RPE and s-IgA after training for HI session 1, 2, and 3; (ii) between TD and s-IgA baseline change for HI session 1; (iii) between RPE and s-IgA baseline change for HI session 2; and (iv) between TD and s-IgA after training for HI session 4 A positive correlation was found for RPE and s-IgA after training for LI session 1
Owen et al. [51] (2015)	23 professional players, 26.8 (±4.6) years, 181.8 (±6.8) cm, 79.3 (±8.1) kg	Total exposure time	HR: time spent at 85 to <90 % HR_{max} (T-HI) and at ≥90 % HR_{max} (T-VHI)	Injury data collection according to methods of Hägglund et al. [70] and Fuller et al. [71]	2 full seasons	A significant correlation was found for T-HI and training injury incidence
Silva et al. [49] (2011)	18 professional players, 25.7 (±4.6) years, 178.1 (±5.7) cm, 76.5 (±9.2) kg	Accumulated match time	No internal load measures	30 m sprint (5 and 30 m sprint times), CMJ, agility T-test, KE and KF isokinetic strength for dominant and non-dominant leg, H/Q, BD, YYIE2	1 full season	Accumulated match time showed significant correlations with changes in physical fitness measures during different periods throughout the season. Positive correlations were found for 5 m sprint time, KE in non-dominant leg, KF in dominant and non-dominant leg, and H/Q in non-dominant leg. A negative correlation was found for H/Q in non-dominant leg
Silva et al. [50] (2014)	14 professional players, 25.7 (±4.6) years, 178.1 (±5.7) cm, 76.5 (±9.2) kg	Accumulated match time	No internal load measures	30 m sprint (5 and 30 m sprint times), CMJ, agility T-test, KE and KF isokinetic strength for dominant and non-dominant leg	1 full season, including the transition period	During the midseason period, accumulated match time significantly correlated with improvements in 30 m sprint time and CMJ

Table 1 continued

Study (year)	Number (n), age [mean (SD)], body height and mass [mean (SD)] of participants	External load indicators	Internal load indicators	Physiological assessment and training outcome, including injuries	Duration	Relationships between load indicators and training outcomes
Thorpe et al. [46] (2015)	10 professional players, 19.1 (±0.6) years, 184 (±7) cm, 75.4 (±7.6) kg	THSR (>14.4 km·h^{-1})	No internal load measures	Daily psychometric questionnaire for player wellness, CMJ, indoor submaximal 5 min cycling/5 min recovery test with HRV and HRR	17 days, during in-season	Perceived ratings of fatigue, HRV, and CMJ were significantly correlated with daily fluctuations in THSR

AU arbitrary units, *BD* bilateral leg strength differences, *CMJ* countermovement jump, CMJ_{as} arm swing CMJ, CMJ_{dl} dominant leg CMJ, CMJ_{ndl} non-dominant leg CMJ, FEHS frequency of efforts at high speed, *HI* high intensity, *H/Q* hamstrings/quadriceps concentric strength ratio, *%HSR* total high-speed running distance as a percentage of total distance covered, *HR* heart rate, HR_{max} maximal HR, *HRR* heart rate recovery, *HRV* heart rate variability, *KE* knee extensor, *KF* knee flexor, *LI* low intensity, *RPE* session rating of perceived exertion, RPE_{mus} session rating of perceived muscular exertion, RPE_{res} session rating of perceived respiratory exertion, *S2* speed corresponding to 2 mmol·L^{-1}, *S4* speed corresponding to 4 mmol·L^{-1}, *SD* standard deviation, *s-IgA* salivary immunoglobulin A, *TD* total distance covered, *T-HI* time spent in high-intensity heart rate zone, *T-VHI* time spent in very high-intensity heart rate zone, *THSR* total high-speed running distance, TL_{RPE} session rating of perceived exertion multiplied by session duration, TL_{RPEmus} session rating of perceived muscular exertion multiplied by session duration, TL_{RPEres} session rating of perceived muscular exertion multiplied by session duration, $TRIMP_i$ individualized training impulse, $\dot{V}O_2$ oxygen uptake, $\dot{V}O_{2max}$ maximal oxygen uptake, *VT* ventilatory threshold, *YYIE2* Yo-Yo intermittent endurance test level 2, *YYIR1* Yo-Yo intermittent recovery test level 1

[49, 50]. Specifically, a relationship was found for players with higher accumulated match time and higher peak torque for extension in the non-dominant leg and for flexion in the dominant and non-dominant leg.

No significant relationship was found between total exposure time and variation in body composition (i.e., estimated body fat, body mass, and fat-free mass) [48].

3.1.2 Relationship with Internal Load Indicators

Five studies examined the relationship between internal load indicators and changes in fitness. Two studies focused on the relationship between RPE and changes in fitness [45, 47], both of which measured RPE in terms of muscular (RPE_{mus}) and respiratory (RPE_{res}) rating of perceived exertion. One inverse relationship was found for an RPE_{mus}-based training load variable and changes in aerobic fitness. It was found that the higher the sustained total RPE_{mus}-based training load (i.e., RPE_{mus} multiplied by session duration), the lower the change in lactate concentration at 13 km·h^{-1} during a submaximal running test [45]. Changes in lactate concentrations for different submaximal running stages were used as a marker for aerobic fitness.

Two studies addressed the effectiveness of high-intensity training using HR-based variables for improving aerobic fitness [42, 43]. It was found that time spent in high-intensity HR zones was related to positive changes in aerobic fitness. In these studies, the high-intensity zone was determined using a progressive treadmill test to define the HR relating to a blood lactate concentration of 4 mmol·L^{-1}. The high-intensity zone in these studies corresponded to a relative HR of 90.2 ± 2.1 % [42] and 91.9 ± 3.0 % [43], respectively. Time spent at high intensity was significantly correlated with improvements in speed at a blood lactate concentration of 2 mmol·L^{-1} ($r = 0.78$–0.84) and 4 mmol·L^{-1} ($r = 0.60$–0.64), in maximal oxygen uptake ($\dot{V}O_{2max}$) ($r = 0.65$), and in Yo-Yo intermittent recovery test level 1 (YYIR1) ($r = 0.66$). Another study focusing on HR has shown a relationship between the individual TRIMP ($TRIMP_i$), an HR-based measure calculated using individual lactate and HR profiles and different measures for aerobic fitness [44]. This $TRIMP_i$ value was calculated for each player for every training session and was associated with improvements in $\dot{V}O_{2max}$ ($r = 0.77$), oxygen uptake ($\dot{V}O_2$) at ventilatory threshold ($r = 0.78$), speed at a blood lactate concentration of 4 mmol·L^{-1} ($r = 0.64$), as well as in YYIR1 ($r = 0.69$). Based on regression analysis, a weekly $TRIMP_i$ of >500 AU was advised to ensure improvements in aerobic fitness during pre-season in professional soccer players.

Los Arcos et al. [47] observed an inverse relationship between total added RPE_{mus} (i.e., the summation of all RPE_{mus} scores for every training session and match) and

15 m sprint times. In addition, players who sustained a higher total RPE_{mus}-based training load or a higher total added RPE_{mus} showed impaired performances on CMJ for both the dominant and non-dominant leg [45].

3.2 Injury and Illness

3.2.1 Relationship with External Load Indicators

One study focused on the relationship between external load indicators and injury. Mallo and Dellal [52] found no relation between total exposure time and incidence of muscle strains. Another study focused on the relationship with illness. This study assessed the relationship between external load variables and salivary IgA as an indicator of immunosuppression in low-intensity and high-intensity training sessions [41]. High-intensity training sessions had significantly higher values for different external load indicators (e.g., distance covered, meterage per minute, distance at high-speed [>21.6 $km \cdot h^{-1}$]) than low-intensity sessions. A higher distance covered for two out of four high-intensity training sessions was related to significant changes in IgA concentrations.

3.2.2 Relationship with Internal Load Indicators

Two studies have assessed the relationship between internal load indicators and injury, and one study focused on the relationship with illness. Two studies examined the relationship between HR and injury. Mallo and Dellal [52] found a relationship between higher mean HR during the season and higher risk of sustaining a muscular strain. In addition, Owen et al. [51] found that the more training time spent in the high-intensity zone from 85 to 89 % of the maximal HR, the higher the injury incidence. This was observed together with increased odds of sustaining an injury during competition for individual players who spent more time at ≥90 % of their maximal HR in training sessions.

In terms of the relationship between internal load and illness, Owen et al. [41] showed a strong inverse relationship ($r = 0.57$–0.72) for RPE values following high-intensity training sessions and salivary IgA testing. This was found in a study examining IgA response after low-intensity and high-intensity training sessions. These low-intensity and high-intensity training sessions differed significantly in RPE values given by players after the training sessions. The results indicated that a higher perceived exertion after high-intensity training sessions is related to a larger drop in salivary IgA, and consequently increased susceptibility for upper respiratory tract infection.

4 Discussion

Our aim was to review the state of knowledge about the connection between indicators for both external and internal load and training outcomes in professional senior male soccer players. In general, this connection is still relatively unexamined in professional soccer. Based on the results of our systematic review, it is evident that only a few studies have examined the relationship between single load indicators and changes in physical fitness, injury, or illness. This observation is remarkable given the abundance of accessible tracking technologies nowadays.

In terms of positive changes in physical fitness, high-intensity activity appears to be related to superior improvements in aerobic fitness. The higher the distance run at high speed (i.e., above 14.4 $km \cdot h^{-1}$), the better the aerobic fitness responses. Additionally, the higher the amount of training time spent in high-intensity HR zones during pre-season, the better the relative changes in aerobic fitness. High-intensity training is indeed considered to be an effective stimulus to improve cardiorespiratory and metabolic fitness [55, 56]. However , it is important to mention that these positive results were only found for a 17-day in-season [46] and pre-season [42, 43] period. Therefore, no definite conclusions are available in terms of the optimal amount of high-intensity training during in-season. The lack of long-term in-season studies focusing on time–motion or HR variables is possibly due to the former FIFA (Fédération Internationale de Football Association) restrictions on wearing electronic performance and tracking systems during official matches, as well as to limitations of HR measures in intermittent team sports [55, 57].

Positive as well as negative relationships were found between exposure time indicators and changes in physical fitness, in particular for power performance including sprinting and CMJ. A higher accumulated match time during a full season was related to improvements in sprinting and CMJ. A rationale for differences between regular starters and non-starters is the specific training stimulus of the match for maintaining or improving physical fitness. A study conducted in elite adult soccer players in the first Croatian junior league [58] showed a positive effect of accumulated match time on speed and jump testing for players who spent more than 1000 min in matches throughout a competitive season compared with players who spent less than 1000 min. Unfortunately, it was unclear if the authors counterbalanced for the missed load in matches for non-starters in terms of additional training sessions. It is possible that players with the highest accumulated match time also had the highest total exposure time, and thus total workload. An overview of the schedule in a winter fixture study over 30 days [59] showed a lower total exposure time for non-starting players, as well as a lower total distance and distance covered in different velocity zones. For non-starting-players, higher workloads preceding a match are considered inappropriate to ensure match readiness in case of a substitution. The combination of a lower accumulated training and match time could be an explanation for the differences in sprinting and jumping between starters and non-starters. This stresses the importance of an optimal workload in order to stabilize or improve power performance for non-starting players throughout a competitive season.

In contrast to this connection between higher in-seasonal accumulated match time and power performance [49, 50], a higher total exposure time during pre-season and the first weeks of in-season was negatively related to relative changes in sprinting and CMJ [45]. This could indicate that training adaptations were temporally suppressed for players with a high total exposure time as a result of functional or non-functional overreaching [60]. An intensified training period, such as a training camp, can cause a temporary decline in physical fitness and can result in an improved physical performance after an appropriate recovery period. Therefore, beneficial training adaptations could have been obtained later on and maintained by regular match play and training sessions, explaining the superior changes in power performance for players with higher accumulated match time during in-season [49, 50].

Additionally, inconsistent findings were apparent for CMJ in two in-season studies. A higher in-seasonal accumulated match time was connected to better CMJ [50], while a higher total exposure time during in-season was related to decreased CMJ [47]. These results could indicate an optimal amount of training load, which is based on an inverted U-curve relationship. This non-linear relationship assumes a turning point between dose (i.e., workload) and response (i.e., changes in fitness) [28]. Interestingly, this relationship is already observed in elite female futsal players for training load and psychophysiological training outcomes [61]. It is worth mentioning that this inverted U-curve for the examined teams could have been different based on their characteristics. The positive relationship was found for a team at the highest professional level [50], while the negative connection was found for a professional third-division team [47].

A negative relationship was found between RPE_{mus}-based measures and relative changes in aerobic fitness [45]. These results revealed an inverse relationship between perceived muscular load and individual changes in aerobic fitness. For RPE and RPE_{res}, no significant associations with changes in aerobic fitness were observed. As far as we know, no single study has shown a clear relation between TL_{RPE} and changes in aerobic fitness. The authors of the study indicated that only 33 % of the variance in aerobic fitness changes could be explained by training load based on RPE_{mus} [45], indicating that the relationship between RPE-based measures and changes in aerobic fitness is still unclear. In addition, perceived muscular load showed a negative effect on individual changes in explosive performance [45, 47]. These results indicate the potential usefulness of this measure to quantify muscular effort using RPE_{mus}. While HR-based parameters appeared to have a positive correlation with changes in aerobic fitness, RPE_{mus} showed negative associations with different physical fitness tests, including tests challenging the musculoskeletal system. This could indicate a dissociation between cardiovascular and musculoskeletal adaptations and recovery time. This observation suggests that for soccer-specific training, the musculoskeletal system needs longer recovery than the cardiovascular system. Indeed, Andersson et al. [62] showed different recovery patterns in neuromuscular and metabolic systems after soccer matches. Based on these findings, it could be advisable to monitor cardiovascular as well as musculoskeletal load to ensure optimal training adaptations and prevent overuse injuries.

In terms of injury, the findings of Owen et al. [51] and Mallo and Dellal [52] showed that more time spent at a higher HR elevates the injury risk. A high HR is the result of high-intensity efforts (e.g., high-speed running or explosive changes of direction) combined with short recovery periods [56]. Indeed, greater amounts of high-speed running (>20 $km \cdot h^{-1}$) were also related to injuries in professional rugby [34]. A lower physical fitness of a player could also be a reason for injuries. Low aerobic fitness, corresponding with a higher relative HR for a similar training load, was associated with a higher injury risk in professional Australian Rules football [63]. Therefore, team training sessions may result in a higher internal load for players with an inferior physical fitness, and consequently a higher amount of fatigue. Acute fatigue could be responsible for injuries during training or matches. However, injuries may also occur as a result of accumulated fatigue in the long-term when multiple intensive workloads do not allow adequate recovery between training sessions or matches.

This review indicates that in professional soccer the relationship between external load and injury for individual players has not been examined longitudinally. However, some evidence is available that points towards similar [64] or increased [9] injury incidence when playing two matches per week instead of one. In other team sports such as rugby and Australian Rules football, time–motion- and accelerometer-derived variables were related to injuries and indeed indicate an increased risk with higher external loads [31, 34, 65]. Additionally, we demonstrate that the relationship between RPE-based variables and injury is unexamined in professional soccer. Again, in other team sports TL_{RPE} has been shown to be related to injury in several trials [35, 36, 66]. Taken together, within professional soccer, external or external: internal load indicators as well as acute: chronic workload ratios could possibly provide a clearer view of the impact of the various locomotor activities that oppose the musculoskeletal system and their relationship with injury. It is important to note that next to its supposed relationship with injuries, continuous monitoring of load is also relevant in quantifying individual responses to training, cumulative fatigue, readiness, and recovery status [67, 68], because these aspects are strongly interrelated with load [3]. This might also be the reason that high-performance practices already apply a wide variety of parameters (e.g., metabolic power and acceleration variables) [69]. However, care should be taken as evidence for the application of monitoring systems is still scarce. Future research should compare different external and internal load indicators concurrently to determine their relevance.

The large diversity of the studies included in this review (e.g., monitoring training sessions and/or matches, duration, team and player characteristics) limits comparability. Some studies have monitored either training sessions or matches, other studies have examined both. Continuous monitoring of all training sessions and matches is a recommendation for future research in order to provide clear insight into the performed workload. Secondly, some of the reviewed studies have focused on a particular period (i.e., pre-season or in-season) and some have examined the training load and outcome relationship over one or more seasons. Therefore, it is difficult to compare results. More research, both short-term (i.e., training camp, pre-season, and in-season period) as well as long-term (i.e. full season) studies, would allow further exploration and provide insight into either short- or long-term relationships between load and training outcomes. Moreover, it should be noted that all studies applied an observational design. As a consequence, environmental factors were not controlled for and the results do not prove causation. Since these studies were conducted in a high-performance environment, sample sizes were relatively small (10–35 players). Therefore, the results should be interpreted with caution and generalizability is limited. Finally, characteristics of the team and players, such as professional level, training history, baseline physical fitness, and injury history, should be taken into account. The reviewed studies were completed at different professional levels, from the highest to the third highest level. In addition, the individual characteristics of the players could limit the external validity of the examined training load indicators and corresponding outcomes. Therefore, in-depth information about individual characteristics is needed to evaluate the applicability for other teams.

5 Conclusions

In professional senior soccer, recent technological developments provide the possibility to measure a variety of different indicators of both external and internal load. This review critically addressed the value of current workload measures in relation to positive or negative training outcomes. Accumulated match and total exposure time appear to be related to positive and negative changes in physical fitness. A higher accumulated match time during the season is linked to better speed and CMJ performance. In contrast, higher total exposure time is related to decreased power performances. Further research is needed that examines the underlying factors involved in total exposure time and, in particular, their connection to changes in physical fitness, injury, and illness. It appears that high-intensive training (i.e., distance covered above 14.4 $km \cdot h^{-1}$ and time spent above >90 % of maximal HR) are effective stimuli to improve aerobic fitness. This type of training involves strenuous activities such as accelerating, sprinting, decelerating, and changing of direction. On the downside, higher volumes of high-intensity training are linked to injuries. That possibly indicates that a higher amount of these efforts entails a greater injury risk, in particular for soft-tissue injury. However, to date, no research in professional soccer has focused on the volume and type of locomotor movements that result in an increased injury risk. Therefore, more detailed research using external load measures to quantify these activities is recommended.

Compliance with Ethical Standards

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